PROJECT UCTF — PAPER 2

Characterizing Universal and Language-Specific Knowledge

What can UCTF safely compress — and what must be preserved?

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TOTAL SENTENCES	KNOWLEDGE CATEGORIES	LANGUAGES	BEST PRESERVATION
240	**6**	**4**	**91.5%**
60 × 6 cat × 4 lang	factual to idiom	EN ZH JA RU	grammar-dependent

1 Research question

Which semantic information is universal across languages and safely compressible by UCTF — and which is language-dependent and must be handled with special care?

Paper 1 established that cross-lingual semantic redundancy exists and is measurable. Paper 2 investigates the nature of that redundancy — categorizing knowledge types and identifying which categories are safe for UCTF compression vs which carry language-specific information that compression would destroy.

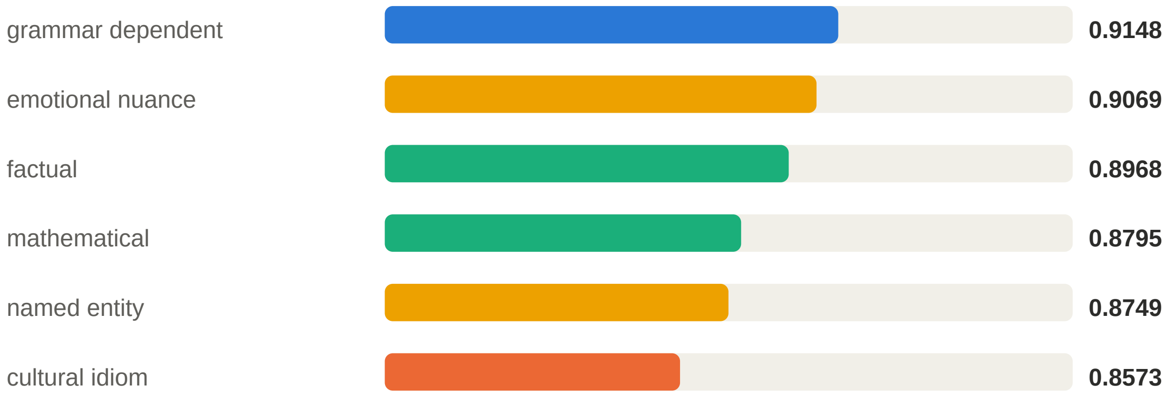
2 Dataset — 6 knowledge categories

60 sentences across 6 knowledge categories, each translated into 4 languages (English, Chinese, Japanese, Russian). 240 total sentences, 360 cross-language pairs per category.

Category	Type	Sentences	Example
factual	Universal	10	Water boils at 100°C.
mathematical	Universal	10	A triangle has three sides.
named_entity	Mixed	10	Einstein was born in Germany.
emotional_nuance	Mixed	10	He laughed to hide his fear.
cultural_idiom	Language-specific	10	It is raining cats and dogs.
grammar_dependent	Lang-structure	10	She gave him a gift yesterday.

3 Main results — similarity by category

Mean cross-language similarity (mE5-small encoder, 6 language pairs):



Category	Mean	Min	Max	Std	Above 0.90	UCTF Safety
grammar_dependent	0.9148	0.8662	0.9615	0.0213	76.7%	■ High
emotional_nuance	0.9069	0.8483	0.9571	0.0244	60.0%	■ High
factual	0.8968	0.8313	0.9432	0.0216	46.7%	■ High
mathematical	0.8795	0.8017	0.9404	0.0309	25.0%	■■ Medium
named_entity	0.8749	0.7893	0.9510	0.0318	16.7%	■■ Medium
cultural_idiom	0.8573	0.7438	0.9748	0.0470	20.0%	■ Caution

4 Key findings

Grammar structure is NOT a compression barrier

Grammar-dependent sentences scored highest (91.5% preserved). mE5 abstracts syntactic structure and captures semantic intent regardless of word order differences.

Cultural idioms are the primary compression risk

Lowest preservation (85.7%), highest variance (std 0.0470), only 58.3% above 0.85. Idiomatic expressions carry language-specific meaning that cannot be safely compressed.

Emotions are more universal than expected

Emotional nuance scored 90.7% — human emotional concepts transfer across language boundaries more reliably than initially hypothesized.

Named entities show moderate divergence

Named entity sentences scored 87.5% with high variance (std 0.0318). Cultural references (Great Wall, Mona Lisa) diverge more than scientific names (Einstein, Amazon River).

Hypothesis partially reversed

Universal categories (factual, math) did NOT score highest. Grammar-dependent and emotional categories outperformed — suggesting mE5 measures semantic intent, not syntactic universality.

5 Most and least universal sentences

Top 5 most universal sentences (highest cross-language similarity):

Category	Similarity	Sentence
grammar dependent	0.9459	She gave him a gift yesterday.
grammar dependent	0.9313	Despite the rain, we continued walking.
emotional nuance	0.9279	He laughed to hide his fear.
grammar dependent	0.9239	If it rains tomorrow, I will stay home.
emotional nuance	0.9233	The joy of reunion after long separation.

Top 5 most divergent sentences (lowest cross-language similarity):

Category	Similarity	Sentence
cultural idiom	0.8103	It is raining cats and dogs.
cultural idiom	0.8287	We will cross that bridge when we come to it.
named entity	0.8426	Leonardo da Vinci painted the Mona Lisa.
cultural idiom	0.8460	He let the cat out of the bag.
named entity	0.8515	The Great Wall of China stretches thousands of kilometres.

6 UCTF design implications

Based on Paper 2 findings, the UCTF encoder design should adopt a tiered compression strategy:

Tier	Categories	Preservation	UCTF Strategy
Tier 1 — Safe	factual, mathematical, grammar, emotion	88–91%	Full compression — low risk of information loss
Tier 2 — Caution	named_entity	87.5%	Compress with entity flagging — preserve proper nouns sepa
Tier 3 — Preserve	cultural_idiom	85.7%	Do not compress — store as language-specific tokens with s

7 Information preservation estimates

Estimated semantic information preserved after UCTF compression (based on mE5 similarity scores):

8 Limitations

• Dataset is manually curated — 10 sentences per category is a small sample.

• Only 4 languages tested — broader language coverage may reveal different patterns.

• mE5-small used — larger encoders (SONAR, mE5-large) may score differently.

• Cultural idiom translations were adapted to target language equivalents — not literal translations.

• Grammar-dependent sentences were constructed to highlight structural differences — may not represent corpus distribution.

• Preservation estimates are proxy measures — actual UCTF information loss requires encoder-decoder experiments in Paper 4.

9 Conclusion

Contrary to initial hypothesis, grammar-dependent sentences are the most safely compressible category (91.5% preservation) — not factual or mathematical content. Cultural idioms represent the highest compression risk (85.7%, std 0.0470) and require special handling in UCTF design. Emotional content is more cross-linguistically universal than expected (90.7%).

These findings directly inform Paper 3 — the formal UCTF design specification — where a tiered compression strategy will be proposed: full compression for universal content, entity-flagged compression for named entities, and language-tagged preservation for cultural idioms.

> *"Grammar structure is not a compression barrier. Cultural idioms are. Emotions are more universal than we assumed. UCTF must be category-aware."*

→ Next — Paper 3

→ Formal UCTF design specification based on Paper 1 and Paper 2 findings.

→ Tiered compression architecture — universal / mixed / language-specific layers.

→ Define the UCTF token format specification.

→ Design encoder requirements for each compression tier.

→ Evaluate SONAR as a candidate encoder for Tier 1 compression.

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